

Homework 3

Due: Feb 25, 2022

All homeworks are due at 11:59 PM on Gradescope.

Please do not include any identifying information about yourself in the handin, including your Banner ID.

Be sure to fully explain your reasoning and show all work for full credit.

LaTeX tips

Here are some more LaTeX commands that might help you

$$\begin{array}{llll} \backslash subseteq & \subseteq & \backslash sqrt{a} & \sqrt{a} \\ & \backslash \{a \mid a \in \mathbb{Z}\} & & \backslash simeq \simeq \end{array}$$

LaTeX *is* required starting with this homework. You should be using the `cs22` document class. Look for a LaTeX guide [here](#).

Problem 1

- a. Find all relations
 - i. from $\{0, 1\}$ to $\{1\}$.
 - ii. from $\{1\}$ to $\{0, 1\}$.
- b. Find all functions
 - i. from $\{0, 1\}$ to $\{1\}$.
 - ii. from $\{1\}$ to $\{0, 1\}$.
- c. Let $S = \{0, 1\}$, $T = \{t \mid t \subseteq S \times S\}$, and R be the set of all possible functions from S to S .
 - i. Can an injection from T to R exist? If so, give one such injection and prove that this mapping is indeed injective. If not, prove why such a mapping cannot exist.
 - ii. Can a surjection from T to R exist? If so, give one such surjection and prove that this mapping is indeed surjective. If not, prove why such a mapping cannot exist.
 - iii. Can a bijection from T to R exist? If so, why? If not, why not?

Problem 2

Determine whether or not each of the following relations is an equivalence relation. Be sure to justify your answers.¹

- a. The relation R on $\mathbb{Z}$ defined by the set of ordered pairs:

$$R = \{(a, b) \mid |a - b| \leq 2\}.$$

- b. The relation R on $\mathbb{R}^2$ defined by the set of ordered pairs:

$$R = \{(a, b) \mid \|a\| = \|b\|\},$$

where $\|a\|$ is the distance from a to the origin in $\mathbb{R}^2$ ($\mathbb{R}^2$ is the set of ordered pairs (x, y) where $x, y \in \mathbb{R}$, also known as the set of points in the plane, and the distance from a point (x, y) to the origin is defined as $\sqrt{x^2 + y^2}$.)

In addition to your proof, answer the following: given a fixed point $p \in \mathbb{R}^2$, the collection of all points related to p gives what familiar geometric object? That is, what is $\{x \mid x R p\}$?

- c. Let $S = \{a, b, c, d\}$. Let R be the relation on S with the graph:

$$\{(a, a), (b, b), (c, c), (a, b), (b, c), (a, c), (b, a), (c, b), (c, a)\}.$$

Problem 3

For International Hot-Drink Day, students propose that the Café has a special where you can try as many flavors of hot-drinks as you want. Each hot-drink comes in regular and mini. You can also not choose to try a particular hot-drink.

Prove, by constructing a bijection², that the total number of possible hot-drink combinations from n hot-drinks is equal to the number of strings of length n containing only characters 'a', 'b', and 'c'.

Example

'abbaa' is a string of length 5 containing only characters 'a', 'b', and 'c'.

¹You might find this resource [here](#) helpful for this question.

²Look for examples [here](#), [here](#), and [here](#). These resources are all linked on the course website!

☕ Problem 4 (Mind Bender — *Extra Credit*)

Mind Benders are extra credit problems intended to be more challenging than usual homework problems and are an exploration into a topic not covered in lecture. This week, we have an exploration into a concept called *equinumerosity* and *countability*.

Don't attempt this unless you feel confident about your solutions for questions 1 through 3!

Thought you learned how to count years ago? Think again!

This question explores how we can use the tools we've learnt, such as functions, to define a more general notion of counting. Especially in the case of counting *infinitely* many things.

Let A and B be sets, not necessarily finite. We say that A and B are *equinumerous* if there exists a bijective function $f : A \rightarrow B$. We use the notation $A \simeq B$ to mean “ A is equinumerous to B .”

Example

$\{a, b, c, d\}$ is equinumerous to $\{0, 1, 2, 3\}$ (what function f can we find?). $\{a, b, c, d\}$ has four elements.

In general, A has n elements if $A \simeq \{0, \dots, n-1\}$. We say that a set is *finite* if it has n elements, for some $n \in \mathbb{N}$.

But what about infinite sets? We say that A is *countably infinite* if $\mathbb{N} \simeq A$ (that is, we can find a bijective function $f : \mathbb{N} \rightarrow A$)³. But equinumerosity of infinite sets can be confusing!

- Show that $\mathbb{N}^+$ (the set $\{1, 2, \dots\}$) is countably infinite: that is, show that there is a bijective function $f : \mathbb{N} \rightarrow \mathbb{N}^+$.

This is not immediately obvious, since $\mathbb{N}^+ \subset \mathbb{N}$! That is, $\mathbb{N}$ contains 1 more element than $\mathbb{N}^+$ but they are still equinumerous!

- Is it true that $\mathbb{Z}$ is countably infinite? Why or why not? If it is, prove so (that is, find a bijection $f : \mathbb{N} \rightarrow \mathbb{Z}$). If it isn't, prove such a bijection doesn't exist.

³Intuitively, this makes a bit of sense. We think of $\mathbb{N}$ to be the counting numbers, so if we can attach a ‘count’ to each element in A , even if A is infinite, it is ‘countable’ in some sense

- c. Show that $\mathbb{N} \times \mathbb{N}$, the set of pairs of natural numbers, is countably infinite.

Hint: Writing down an explicit bijection f is hard. You can draw a picture here, or explain precisely how you would “count” all of the pairs.

- d. We give you these two facts, that you do not need to prove yourself:
- i. The relation $\simeq$ is transitive: if $A \simeq B$ and $B \simeq C$, then $A \simeq C$.
 - ii. If $A \subseteq B$, A is not finite, and B is countably infinite, then A is countably infinite.

Using the above two facts and part c., show that the set of positive rational numbers $\mathbb{Q}^+$ is countably infinite. Can you also conclude that $\mathbb{Q}$ is also countably infinite?

- e. It’s starting to sound like a lot of sets are countably infinite! Can you think of an infinite set that is *not* countable? You don’t need to prove that it is uncountable, but indicate why you think it isn’t.

The following aren’t for credit, but are some extra food for thought:



- f. What set did you come up with? Can you prove that it is somehow not countably infinite?



- g. You might wonder whether *all* uncountably infinite sets are equinumerous? Is there anything *even bigger* than uncountable infinity? Can you think of two uncountably infinite sets that aren’t equinumerous. How would you prove so?

You might want to look up *diagonalization* and *Cantor’s Theorem*.